

Mohammad Alvi Refat

[Website](#)[Scholar](#)[GitHub](#)moref17@gmail.com

EDUCATION

The Graduate Center, City University of New York

Sep 2025

M.S. in Astrophysics

Advisor: Dr. Lucy Lu

Thesis: Starspot Inference Using Light Curve Inversion Techniques

CUNY Baccalaureate for Unique and Interdisciplinary Studies

Jun 2022

B.S. in Computational Astrophysics, *cum laude*

RESEARCH INTERESTS

Computational materials physics; computational biophysics; complex systems.

RESEARCH EXPERIENCE

Starspot Inference with Light Curve Inversion Techniques [\[Thesis\]](#)

American Museum of Natural History

Aug 2023 – Sep 2025

M.S. Thesis Researcher | Advisor: Dr. Lucy Lu

- Implemented forward- and inverse-modeling workflows with **fleck** and **starry** to analyze 1,000 synthetic, evolving starspot light curves generated with **butterpy**.
- Evaluated recovery of stellar rotation, activity indicators, and surface structure across variations in photometric noise, viewing inclination, and spot configuration.
- Recovered rotation and statistical activity indicators in the tested light curves, while finding that individual starspot locations and surface maps were not uniquely constrained.

Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres [\[Code\]](#)

American Museum of Natural History

May 2021 – Aug 2023

BDNYC Researcher | Advisor: Dr. Johanna Vos

- Applied spherical-harmonic surface mapping with **starry** to three-dimensional atmospheric simulations and 20-hour Spitzer light curves of two isolated planetary-mass objects.
- Evaluated model complexity, identifying spherical-harmonic degree $l = 5$ as sufficient to reproduce the relevant light-curve structure.
- Identified atmospheric evolution as a limitation of static-map fits, motivating rotation-by-rotation mapping of changing cloud structures.

Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2 [\[Code\]](#) [\[Paper\]](#)

Sloan Digital Sky Survey Faculty and Student Team

May 2020 – May 2021

SDSS FAST Student Researcher | Advisor: Dr. Allyson Sheffield

- Combined APOGEE-2 spectroscopy, Gaia astrometry, stellar kinematics, and chemical abundances to identify candidate Jhelum stream members in a sample of 289 red giants.
- Contributed to identifying a new metal-poor red-giant member and compared its chemical and orbital properties with candidate Gaia–Enceladus–Sausage stars.
- Coauthored a peer-reviewed article in *The Astrophysical Journal* on the stream’s membership, chemical properties, and progenitor origin.

Looking at Star Formation Through Chemistry

American Museum of Natural History

Aug 2018 – May 2020

AstroCom NYC Scholar | Advisor: Dr. Allyson Sheffield

- Processed and normalized high-resolution echelle spectra with **SMHr**, measured absorption-line equivalent widths, and iteratively fitted stellar atmospheric parameters and elemental abundances for chemical-tagging analysis.
- Validated the workflow against published Arcturus measurements, recovering effective temperature within 50 K and metallicity within 0.07 dex of the reference values.

Two-Point Statistics in the Star-Forming Interstellar Medium [\[Code\]](#)

Flatiron Institute, Center for Computational Astrophysics

Jun 2018 – Aug 2018

AstroCom NYC Scholar | Advisor: Dr. Chang-Goo Kim

- Developed Python routines to compute and visualize two-point metallicity correlations in TIGRESS magnetohydrodynamic simulations of the star-forming interstellar medium.
- Measured a characteristic metallicity half-correlation length of 120 ± 25 pc and estimated a turbulent metal-mixing timescale of ~ 10 Myr.

PUBLICATIONS

Peer-Reviewed Publications

Sheffield, A. A., A. Z. Subrahimovic, **M. Refat**, et al. (2021). “Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2”. *The Astrophysical Journal* 913.1, p. 39. [DOI](#).

Thesis

Refat, M. (2025). “[Starspot Inference Using Light Curve Inversion Techniques](#)”. M.S. Thesis. CUNY Graduate Center.

ACADEMIC PRESENTATIONS

Talks

“Star Spot Inference Using Light Curve Inversion Techniques” <i>CUNY Masters Graduation</i> — Simons Foundation	2025
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>240th American Astronomical Society Meeting</i> — Pasadena, CA	2022
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>Center for Computational Astrophysics Symposium</i> — Simons Foundation	2021
“Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2” <i>American Museum of Natural History REU Symposium</i> — New York, NY	2020
“Where Were Stars Born in the Milky Way?” <i>American Museum of Natural History REU Symposium</i> — New York, NY	2019
“Two-Point Statistics in the Star-Forming ISM” <i>Center for Computational Astrophysics REU Symposium</i> — Simons Foundation	2018

Posters

“Starspot Inference with Light Curve Inversion Techniques” <i>245th American Astronomical Society Meeting</i> — National Harbor, MD	2025
“Star Spots with Starry” <i>Cool Stars 22</i> — University of California San Diego	2024
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>Center for Computational Astrophysics Gotham Fest</i> — Simons Foundation	2023
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>241st American Astronomical Society Meeting</i> — Seattle, WA	2023
“Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres” <i>Center for Computational Astrophysics Gotham Fest</i> — Simons Foundation	2021
“Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2” <i>237th American Astronomical Society Meeting</i> — Virtual	2021
“Where Were Stars Born in the Milky Way?” <i>Center for Computational Astrophysics Gotham Fest</i> — New York University	2019

PROFESSIONAL APPOINTMENTS

Adjunct Lecturer <i>Baruch College, City University of New York</i>	<i>Aug 2025 – May 2026</i>
– Delivered introductory physics lecture and laboratory instruction for PHYS 2002, PHYS 2003, and PHYS 3001. – Designed lectures, laboratory exercises, and assessments emphasizing experimental methods, computational problem solving, data analysis, and uncertainty evaluation.	

COURSES TAUGHT

PHYS 2002 – Introductory Physics Lecture	Fall 2025, Spring 2026
PHYS 2003 – Introductory Physics Laboratory	Fall 2025
PHYS 3001 – Physics II Laboratory	Spring 2026

TECHNICAL SKILLS

Programming Languages	Python, MATLAB, Bash, Java
Computational Methods	Inverse problems, Bayesian inference, numerical optimization, time-series and spectral analysis, spatial statistics
Scientific Computing	NumPy, SciPy, Pandas, Astropy, Lightkurve, Astroquery
Visualization	Matplotlib, Glue, Manim
Research Tools	Git, Linux, Jupyter Notebook, L ^A T _E X

OUTREACH

Science Research Mentoring Program Meeting — Speaker	2021
Science Research Mentoring Program Family Orientation — Panelist	2019, 2020